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Experimentally validated numerical models to assess tsunami hydrodynamic force on an elevated structure

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ABSTRACT

Recently, the North American and Japanese authorities began combining the tsunami forces with other loads in their structural design guidelines. Nonetheless, due to the infrequent nature of tsunamis, the provisions may benefit from complementary insights on the qualitative and quantitative characterization of the extreme phenomena and their interaction with coastal structures. The goal of this paper is to explore reliable and relatively accessible computational techniques to determine pressures and forces due to tsunami-like waves on elevated structures. Hydrodynamic flow quantities are simulated for the tsunami-like waves using an Eulerian scheme (Shallow Water equations solved by the Finite Volume method), a Lagrangian scheme (Navier–Stokes equations solved by the Smoothed Particle Hydrodynamics method), and a coupled Eulerian Lagrangian modeling approach. The numerical solutions are validated against experimental data acquired from an experimental campaign performed at the large wave flume of the Hinsdale Wave Research Laboratory, Oregon State University. The correlations between the experimental data and the numerical solutions highlight the advantages and disadvantages of the simulation techniques used, contributing to increase the confidence levels in their use in determining tsunami forces for use in structural assessments and design.

1. Introduction

Extreme and catastrophic tsunamis, such as 2004, Sumatra, Indonesia, and 2011, Tohoku, Japan, highlighted the need for mitigating their impact on the built infrastructures. In addition, reconnaissance efforts and analysis thereof emphasized the need of a more accurate definition of tsunami loads on structures [1,2].

Propelled by the lessons learned, two countries began to develop tsunami guidelines for structural design. In Japan, the Ministry of Land, Infrastructure, Transport and Tourism published guidelines that include provisions for determining the tsunami as equivalent hydrostatic lateral loads, function of the inundation depth taken directly from tsunami inundation maps provided by coastal municipalities [3]. In the United States of America, a new chapter 6 dedicated to tsunami design loading was included in the 2016 edition of the ASCE 7 [4]. ASCE 7 defines use of hydrostatic and hydrodynamic tsunami effects, where the tsunami hydrostatic effect is intrinsic to inundation depth, while the hydrodynamic effect is characterized by the product of the inundation depth by the squared velocity, known as the linear momentum flux [2,5]. ASCE 7 recommends two possible approaches for evaluating the momentum flux. First, the Energy Grade Line Analysis, which makes use of 1 *in* 2500-year probabilistic tsunami design maps that were developed for the US western states of Alaska, California, Hawaii, Oregon, and Washington. Alternatively, site-specific tsunami hazard analysis using numeric tools to compute the tsunami inundation depth and flow velocity can be prescribed, requiring tsunami generation, propagation, and inundation modeling [6–9]. The tsunami simulations of generation and open-ocean propagation phases require an ability to characterize the response at macro-scales, accepting linear equations for a hydrostatic approach as governing equations, such as Shallow Water (SW) equations. The inundation phase requires complementary high-resolution topo-bathymetric data and more sophisticated governing equations, such as Navier–Stokes (NS), Reynolds Averaged Navier–Stokes (RANS), Boussinesq or non-linear SW equations, all integrated in depth and

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taking into account the variations in the vertical direction and the physics of the environment.

Classic Eulerian discretized methods of Finite Differences (FD) [6,7] or Finite Volumes (FV) [10,11] are valid numerical schemes to assess tsunami hydrodynamic quantities. Nonetheless, Eulerian mesh-based methods have inherent limitations in the treatment of complex flows and geometries, requiring mathematical transformations that represent additional computational expenses. Additionally, the precise locations of the free surfaces under large deformation, deformable boundaries and moving interfaces within the frame of the fixed Eulerian grid are difficult to handle. Moreover, the eligibility of SW governing equations is only guaranteed if the wave height is significantly smaller than the wave length, allowing to neglect the modeling of the dispersive component of the wave(s). Otherwise, the hyperbolic system of governing equations is not capable to reproduce the hydraulic phenomena inherent to the flow.

The Lagrangian mesh-less Smoothed Particle Hydrodynamics (SPH) method offers several advantages with respect to Eulerian methods when tackling fluid impact or interaction with structures, being capable of reproducing complex fluid flows, highly non-linear free-surface phenomena, conveniently treating large deformations and handling variable domains in time, such as piston-type wave-makers, making SPH a promising tool to use in the characterization of tsunami-like waves [12–14].

Recent works on the characterization of hydraulic phenomena, such as regular, non-regular, solitary, and tsunami waves [15–24], have demonstrated the capacities of the NS-SPH numerical scheme to reproduce physical phenomena of fluids and their interaction with other fluids [25] and/or solids, which makes SPH particularly interesting to use in the framework of coastal engineering research. Its suitability to model the impact and over-topping effects on vertical structures has been explored [26–32], but comparatively less attention has been devoted to studying the interaction of fluids with modified [33,34] or elevated structures [35–37]. In shallow coastal areas, the impact of tsunami-like waves can also have an important vertical hydrodynamic component, which threatens elevated structures such as piloti-type buildings, bridges, wharves with open typology, among others.

However, the SPH method suffers from numerical diffusion, has a low convergence rate, needs additional algorithms to model turbulence and presents large pressure fluctuations due to spurious pressure waves in the untreated weakly compressible formulation [38]. Since SPH is a relatively new tool for engineering purposes, additional work is needed to demonstrate the advantages of SPH over the well-established classic mesh-based methods [39] that are computationally cheaper [40].

Recent hybrid techniques have emerged to overcome the inherent limitations associated to the SPH computational costs [41–45]. By solving the linear(-ish) part of the domain using the comparatively faster but smoother mesh-based approaches, the SPH domain governed by sophisticated systems of equations is reduced, decreasing the computational expense and maintaining the level of detail of the numerical solutions. However, the algorithms of hybridization tend to be complex.

In addition, there is a lack of systematic validations of the SPH numerical solutions using detailed experimental measurements of hydrodynamic flow conditions as well as horizontal and vertical pressures and forces on the infrastructure.

Aiming to contribute to the investigation of the tsunami-like wave loading on structures in both horizontal and vertical directions, the objective of this paper is to present and benchmark numerical approaches to characterize the impact of broken and unbroken waves on an elevated structure that can be obtained using low-cost computation demand and accessible local computing clusters. In this paper, three numerical approaches are described and their use exemplified. The first numerical approach is a purely numerical characterization of the tsunami flow and impact exerted on the elevated structure using a Lagrangian SPH-based code [40] to solve the NS equations to compute flow depth, velocity, pressures and forces. In the second approach, the hydrodynamic quantities of the tsunami-like waves are assessed using an Eulerian FV-based code, implemented by four of the co-authors [11,46], which uses the FV method to solve the non-linear SW equations. Based on the numerical solutions of the flow quantities, analytical equations are then used to assess the hydrodynamic forces. A brief incursion on the drag coefficient (C_d) used in the analytic equations is discussed. The third numerical approach couples the SW-FV and NS-SPH schemes whereby the Eulerian–Lagrangian coupling is performed by means of Dirichlet boundary conditions prescribed by average operators. The three numerical approaches are benchmarked against laboratory data of a recent experimental campaign performed at the large wave flume at Hinsdale Wave Research Laboratory, Oregon State University [47–49]. The correlations between experimental and numerical solutions highlight the potential of each numerical scheme, the sophisticated but computationally expensive nature of NS-SPH and the fast but smooth SW-FV solutions, while the coupled numerical configuration is regarded as a compromise between both.

2. Experimental data

An experimental campaign conducted at the large wave flume at Hinsdale Wave Research Laboratory, Oregon State University was carried out to provide laboratory data to assess the impact of waves on elevated structures. Detailed information on the experimental campaign can be found in [47–49].

The experiment consisted in the generation of tsunami-like waves, which propagated in a variable sloping tank, and interacted with an elevated structure. By varying the still water level and the piston velocity, unbroken and broken (bore) long-waves were generated. The objective of the experiment was to develop a dataset of measures concerning: (1) water height and velocity due to unbroken and broken (bore) tsunami-like waves, and (2) forces and pressures exerted on an elevated structure resembling a scaled two-story building. The experiment was instrumentally monitored and physical data was recorded.

Based on the experimental results, the performance of the numerical modeling approaches can be assessed for nonturbulent unbroken and broken wave cases, as well as cases without and with an air gap between the still water level and the bottom of the test specimen.

Sections 2.1 and 2.2 detail, respectively, the experimental setup and the quantification of the drag coefficient on quasi-steady and bore-like flows, in both the horizontal and vertical directions.

2.1. Experimental setup

Fig. 1(a) depicts the setup, including an elevation and a plan view of the flume with the instrumentation. The length of the flume is 87.430 m, from the mid position of the wave-maker paddle to the end, the width is 3.660 m, and the depth is 4.572 m. Flat and sloping conditions constitute the bathymetry of the channel. These include (1) a flat section from 0 m to 14.250 m, (2) a slab elevated 0.150 m from the flume base between 14.250 m and 17.907 m, (3) a 1/12 slope with a projected horizontal length of 10.971 m followed by (4) a second 1/24 slope with a projected horizontal length of 14.628 m, (5) a flat extension of 36.570 m, and (6) a final 1/12 slope with projected horizontal length of 7.354 m.

The flume is equipped with a piston-type wave-maker with a hydraulic actuator assembly of 4.2 m maximum stroke and 4 m/s maximum velocity. The wave-maker is capable of producing regular, irregular and solitary waves, with periods ranging between 0.8 s to 12^{+} s. More information on the facility can be found at https://wave.oregonstate.edu/large-wave-flume. The at-rest position of the piston defines the null state of the coordinate system. The initial position of the paddle is −2.060 m. The piston operated under displacement control, for the generation of unbroken and broken long-waves. The still water level was set at $z = 2.00$ m for the unbroken wave case and $z = 1.85$ m for the broken case. The two waves cases were thus quite distinct. The shape of the unbroken wave was nonturbulent, with an approximate duration

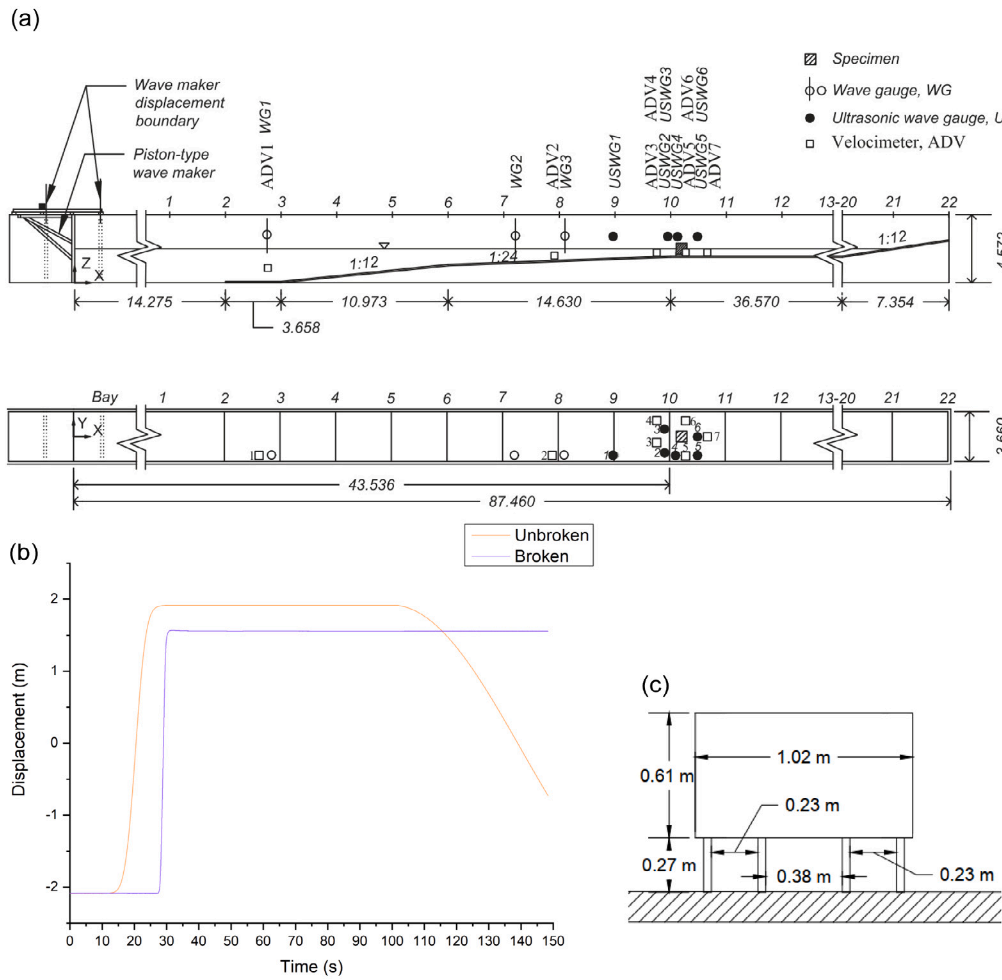


Fig. 1. Overview of the experimental campaign: (a) elevation (top) and plan views (bottom), (b) time–histories of piston displacement for unbroken and broken waves, and (c) geometry of the elevated structure specimen (frontal view).

of 25 s, average height of 30 cm and 1.5 ms^{-1} velocity. The shape of the broken wave was steep and abrupt, with half the duration, circa of 1 $m\pm 20$ cm height and the velocity measuring around 2.5 ms^{-1}. Fig. 1(b) depicts the horizontal displacements of the wave-maker.

An elevated structure resembling a two-story building supported by piles was implemented in the model. The scaled structure was a 1.016 m square in plan with a height of 0.615 m, supported by twelve 0.010 m-square columns of 0.264 m-height. The center of the base of the flat bottom surface of the structure was positioned at $x = 44.266$ m, $y = 0.002$ m and $z = 2.002$ m. Fig. 1(c) depicts the front view geometry of the elevated structure.

The instrumentation of the experimental campaign included three wage gauges (WG) and five ultrasonic wave gauges (USWG) to record the water free-surface along the flume and surrounding the structure, respectively. Seven acoustic doppler velocimeters (ADV) recorded the velocities along the flume and near the structure. Nine load cells (LC) measured stream-wise, transverse, and vertical wave forces. Fourteen pressure gauges (PG) recorded the horizontal and vertical pressure exerted against the structure using three different layout dispositions to map the pressure distributions on the structure. The tests and data acquisition were repeatedly trialed.

Considering the large amount of data involved, the subsequent experimental–numerical validation considers one record of each type of measure, including USGW6, ADV3 to ADV7, for water elevation and flow velocity, respectively. In addition, the stream-wise force was measured by load cell LC5, while the total vertical forces were measured as the sum of forces measured by LC6 to LC9 sensors and these are used for correlation of the numerically obtained horizontal and vertical forces with the experimental results. Lastly, horizontal and vertical pressures are compared for PG4 and PG11 sensors and corresponding locations, respectively. The experimental data is characterized as the average value of all trials complemented with the respective standard deviation. The average data concerning the waves elevation, pressures, and forces, present very small deviations in the measurements acquired in the different trials. However, the inherent difficulty of velocity data acquisition partially constrained the quality of the correlations that were possible. As applying filters often leads to lose information detail, it was considered preferable to maintain the use of unfiltered laboratorial records in the benchmark process.

2.2. Drag coefficient

In fluid mechanics, the hydrodynamic force is the product of the momentum flux affected by a dimensionless coefficient known as drag coefficient, C_d, that represents the behavior of a quasi-steady flow against a fully submerged solid. In tsunami literature, it also applies to partially submerged objects and moderate to high tsunami velocity [50, 51].

Over the years, extensive and numerous physical experiments were performed to assess the influence of several parameters, such as the shape of the object, bore height, Reynolds number characterizing laminar or turbulent flow, Froude number characterizing sub- or supercritical flow, ratio between water depth and object width, and blockage coefficient considering the width of the tank and of the column. Most authors agree on the sensitivity of C_d and on the difficulty to calibrate it [5,50,52–60], demonstrating that C_d is not a function of any single non-dimensional grouping, but strongly depends of a combination including the Froude number, ratio between the widths of the propagation channel and of the obstacle, and the shape of the object.

Aiming to contribute for the formulation of gravity-controlled free-surface flow, the C_d of tsunami-like unbroken and broken waves traveling over an inundated bed is briefly tackled. Eq. (1) is used to quantify C_d values, which are obtained by regression to best-fit the peak of the first wave, where the most interesting and important results for the forces and pressures exerted on the structure occur.

$$F_{dynamic} = \frac{1}{2}\,\rho\, C_d\, B\,(hu^2)_{max} \tag{1}$$

where $\rho = 1000\ \mathrm{kg/m^3}$ is the density of the fluid, C_d is the drag coefficient, $B = 1.016\,\mathrm{m}$ the characteristic width of the solid subjected to the action of the fluid and hu^2 is the momentum flux. The hu^2 was calculated using records of USWG6 and ADV7 instruments for the stream-wise direction, while records from ADV3 to ADV6 were considered for calculation in the vertical direction.

Based on the analysis of the experimental data and fitting to Eq. (6), the coefficient in the horizontal direction was valued in $C_d = 1.1$, for the unbroken wave case, and $C_d = 1.3$, for the broken wave case. The values are lower than the provisions from the ASCE 7 standard, Table 6.10-2, where $C_d = 2$ recommendation depends on the ratios of width to depth and width to height of the submerged surface area. The numerical undervaluation of the lateral C_d is reasonable due to the elevated geometry of the obstacle. Regarding the values in the literature, the C_d value is in agreement with some of the studies performed for rectangular-shaped columns [60,61] and studies involving circular-shaped [62–64] that logically induce smaller values of resistance to the flow than pointy objects; while it is underestimated when compared to other studies on rectangular columns whose values range from $C_d = 2$ to $C_d = 4$ [63,65,66].

In the vertical direction, for the unbroken wave case, the obtained $C_d = 2.0$ is smaller than the recommendation of the Federal Emergency Management Agency (FEMA) guidelines ($C_d = 3.0$) [67]. For the broken wave case, the greater turbulence of the flow and magnitude of the stream-wise force promoted a significant reduction of the uplifting force that is the result of the sum between front and back loads with opposite signs. The value of $C_d = 0.2$ is similar to under-unity values assessed for aerodynamic problems, reinforcing the assumption that the hydrodynamic force equation recommended by FEMA [67] and ASCE [4] adopting a static drag coefficient is relatively adequate to evaluate the net fluid force on a body in a quasi-steady flow [50,51], but inadequate to estimate the hydrodynamic load when the tsunami flow makes the transition from laminar to turbulent flow, usually at Reynold's number of the order of $Re \approx 10^5$.

Fig. 2 illustrates the potential time dependency influence when defining the tsunami design loading. The semi-analytic formulation (adequate to determine peak values) is adopted to calculate time-series for the force exerted on an elevated structure, using the hydrodynamic quantities measured by the laboratorial instruments, USWGs and ADVs, and the C_d values previously quantified for peak values of the unbroken wave case. The curves are compared to recorded forces measured by LC5, for the horizontal directions, and the sum of LC6 to LC9, for the vertical direction.

As expected, the fitting between recorded and calculated curves displays a good agreement for the first wave since the C_d was calibrated by regression to reproduce the peak of the first wave. For the horizontal direction, the variations after the first wave may be acceptable to design lateral tsunami-resistant systems. Conversely, a significant underestimation of the vertical force exerted on the bottom of the structure occurs, partially neglecting its intensity over time. As depicted in Fig. 2, the hydrodynamic effect in the vertical direction (at right) has an important expression over time, with the peaks of the first four waves being greater than the maximum peak value in the horizontal direction (at left). The optimistic characterization of the tsunami hydrodynamic effect in the vertical direction represents a strong potential contribution to under-design structural resisting systems of elevated structures. Moreover, Fig. 2 only represents the more predictable nonturbulent unbroken wave case. A brief incursion on the broken wave case, adopting Eq. (1), hydrodynamic quantities instrumentally recorded near the structure and $C_d = 3.0$, demonstrate less accentuated waves following the first bore. Adopting a slightly conservative value of C_d potentially represents a procedure to avoid a dynamic characterization of the tsunami hydrodynamic effects in the vertical direction, simplifying the structural design for such cases.

3. Numerical data

The assessment of the hydrodynamic effects of the tsunami-like wave impacting the elevated structure is performed considering three distinct numerical configurations:

- built-in numerical computation on Lagrangian NS system using the SPH framework implemented in DualSPHysics [40];
- semi-analytic quantification using the computation of the tsunami-like hydrodynamic quantities by Eulerian discretization scheme that considers the SW equations in a fixed reference system, solved in FV-based scheme implemented using a C++ code [11,46]. The force exerted on the structure is calculated using the hydrodynamic equation, Eq. (1);
- coupled Eulerian SW-FV and Lagrangian NS-SPH to simulate the tsunami propagation and inundation phases, respectively.

Fig. 3 depicts the numerical configurations used to perform the simulations, including two different solvers (SPH and FV) and three different Dirichlet boundary conditions, which are transmissive (TBC), for the FV, and moving (MBC) and open (OBC), for the SPH. The symbols Y, for "Yes", and N, for "No" represent the models using a MBC resembling the physical displacement of the piston or the models using TBC imposing hydrodynamic quantities, respectively.

Sections 3.1 and 3.2 describe, respectively, the numerical schemes used to assemble the configurations and the modeling conditions prescribed to perform the numerical simulations.

3.1. Numerical schemes

3.1.1. Built-in force computation: Lagrangian model and SPH method

Fluids are typically governed by the partial differential NS equations, derived from three conservation principles (of mass, momentum and energy), describing the motion of a continuum fluid. Assuming the fluid is incompressible [68,69], the energy equation is dropped since the pressure derives from the free divergence constraint, $\nabla \cdot V = 0$. Traditional SPH methods are explicit in time. However, incompressible fluids require the resolution of a Poisson equation to evaluate the correct pressure that preserves the divergence free property. To avoid supplementary pressure equations, a popular technique is often used assuming the fluid to be weakly compressible (function of density, and solvable by Tait's relation for the state equation), leading to a strictly hyperbolic system (at least when the diffusion term is canceled). The open-source DualSPHysics software [40] is capable of solving both incompressible and weakly compressible fluids. In the present study, the DualSPHysics software (version 5.0) is used to solve the Lagrangian system of NS equations for weakly compressible fluids complemented by Tait's state equation (Eq. (2)):

$$\begin{cases} \dfrac{dV}{dt} = -\dfrac{1}{\rho}\nabla p + g + \Gamma \\ \dfrac{dr}{dt} = V \\ \dfrac{d\rho}{dt} = -\rho\nabla V \\ P = \beta\left[\left(\dfrac{\rho}{\rho_0}\right)^{\gamma} - 1\right] \end{cases} \tag{2}$$

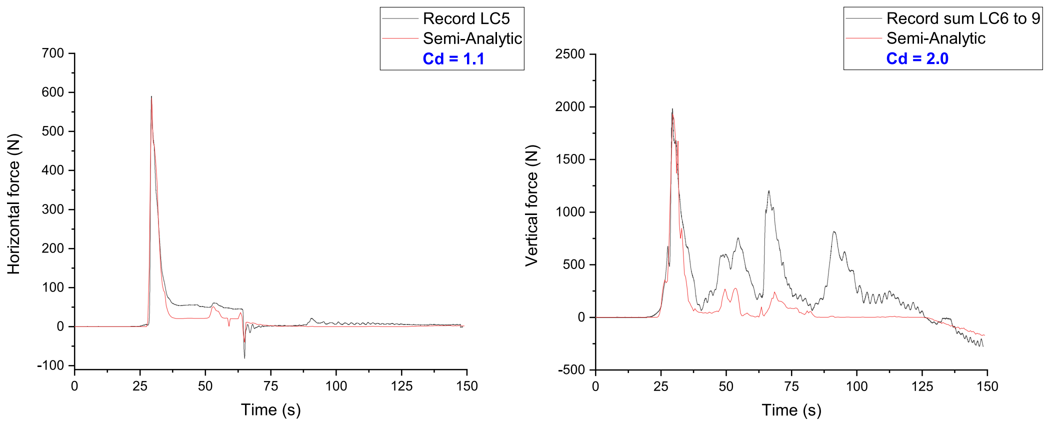


Fig. 2. Comparison of recorded and calculated forces exerted on the structure, considering stationary C_d estimated to fit the peak of the wave in the horizontal (left) and vertical (right) directions. Example considering the unbroken wave case.

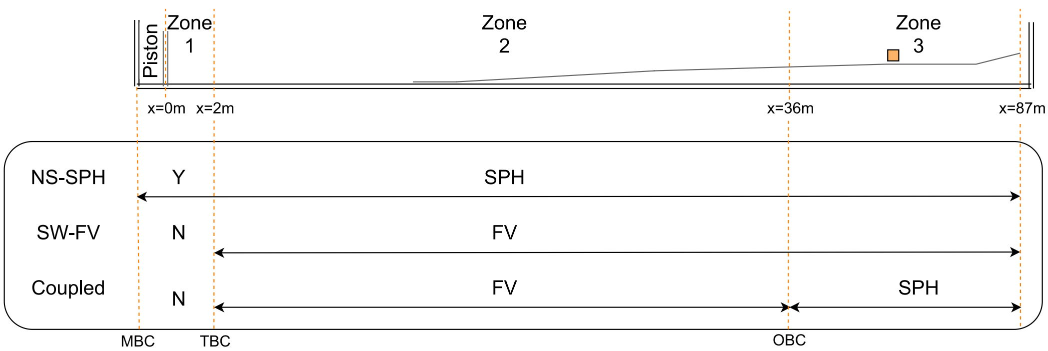


Fig. 3. Schematic summary of the numerical protocols, representing the domains where the SPH and FV solvers are used and the Dirichlet boundary conditions implemented, including transmissive (TBC) for the FV solver, moving (MBC) and open (OBC) for the SPH solver.

where $V = (u, v, w)^T$ is the velocity field, t is the time, p and ρ are pressure and density, respectively, g is the gravitational acceleration, r is the position, and Γ represents the dissipative term. In Tait's equation, density and pressure are related assuming the reference density ρ_0 equals the gas density at rest, while parameter $\gamma = 7$ is the polytrophic constant and $\beta = \rho_0 C_s^2/\gamma$, where C_s is the speed of sound.

To solve system of Eq. (2), the SPH method substitutes the continuum properties of the fluid by a set of discrete interacting particles. Each particle is defined by its fluid properties (density, velocity, pressure) and related to the neighboring particles by means of a kernel function W to mimic the integro-differential operators. Any function, f, given with respect to the particle position vector, r, is approximated through the integration of the relevant properties of all the particles that lie within the radius of the smoothing length, as given by Eq. (3):

$$\langle f(r)\rangle = \int_{\Omega} f(r')W(r-r', h_{sph})dr' \tag{3}$$

where $W(r-r')$ is the kernel, h_{sph} the smoothing length defining the radius of influence of domain Ω and $\langle\ldots\rangle$ an approximation operator.

Considering a discrete particle set, Eq. (3) is substituted by a discrete summation over all the particles inside the kernel. Consequently, for any particle a with position $r_a = r_a(t)$, the discrete average is given by Eq. (4):

$$f(r_a) = \sum_{b\in N(a)} f(r_b)\frac{m_b}{\rho_b}W(r_a - r_b, h_{sph}) \tag{4}$$

where m is the mass and b stands for the particles that belong to the discrete kernel support of particle a, $N(a)$. Integrating Eq. (2) by parts (divergence theorem) and assuming that the kernel support is compact, the derivatives are now supported by the kernel function and the conservation equations of momentum and mass continuity, complemented by Tait's state equation, are now given by Eq. (5):

$$\begin{cases} \dfrac{dv_a}{dt} = -\displaystyle\sum_{b\in N(b)} m_b\left(\frac{p_a+p_b}{\rho_a\cdot\rho_b} + \prod_{ab}\right)\nabla_a W_{ab} + g \\ \dfrac{dr_a}{dt} = v_a \\ \dfrac{d\rho_a}{dt} = \displaystyle\sum_{b\in N(b)} m_b v_{ab}\nabla_a W_{ab} \\ P_a = \beta\left[\left(\dfrac{\rho_a}{\rho_0}\right)^{\gamma} - 1\right] \end{cases} \tag{5}$$

where $\prod$ is an additional artificial viscosity term introduced for the sake of stability [70].

The computation of the force exerted by the fluid against a solid object at time t^n is computed as the summation of the momentum variation of each particle in the neighborhood of the solid (Eq. (6)):

$$F = -m_k\sum_{k=1}^{n}\left(\frac{dv_k}{dt}\right) \tag{6}$$

where k refers to the boundary particles that form the solid, m_k is the mass and $\frac{dv_k}{dt}$ is the acceleration computed for time-step t. Note that the summation is only carried out for the particles that hit the surface of the solid in the time interval $[t^n, t^{n+1}]$, by solving the momentum equation taking the distance of interpolation $<$ 2 h [31,71].

The fluid–structure interaction can lead to large free surface deformation and wave breaking. The mesh-free Lagrangian nature of the SPH method confers suitability to model complex flows with interfaces and multiple phases [38], while the Navier–Stokes equations naturally provide detailed flow characterization [72]. Some authors raised awareness for the importance of the solid boundary treatment surrounding the structure [73] and air-phase modeling that influence the numerical predictions of the fluid motion and structural response, consequently influencing the assessment of pressure and force on the structure [72]. The formulation in [25] is often used to modify Tait's equation of state and account for the properties of the different phases by adding a constant background pressure and cohesion forces between the particles of a single phase to model two-phase water–air interface. However, modeling a second phase increases the computational complexity and time [74]. As the goal of the present study is to maintain relatively cheap computational costs, and the multi-phase modeling is primarily important in water-compressible air flows, the simulations herein performed neglect the multi-phase modeling of air–water interface.

3.1.2. Semi-analytic formulation: Eulerian model and FV method

Based on Bernoulli's Principle, which is related to the conservation of energy, the pressure, flow height and velocity are correlated such that the sum of internal, kinetic and potential energy remains constant along a streamline of the fluid. The potential energy is directly related to the hydrostatic component of the force, where as the kinetic energy relates to the hydrodynamic component of the force. The peak hydrodynamic force is derived from the maximum flux momentum and is given by Eq. (1). The hydrodynamic quantities required to define the momentum flux, wave elevation and flow velocity, are computed by the Eulerian FV scheme that solves the system of SW equations.

The SW equations (Eq. (7)) derive from a depth-integration of the incompressible NS equations and represent an eligible approximation of the NS equations when the horizontal scale characterized by the wavelength is much larger than the vertical scale representative of the depth. One has,

$$\begin{cases} \partial_t h + \partial_x(hu) + \partial_y(hv) = 0 \\ \partial_t(hu) + \partial_x(hu^2 + \frac{g}{2}h^2) + \partial_y(huv) = -gh\partial_x b \\ \partial_t(hv) + \partial_x(huv) + \partial_y(hv^2 + \frac{g}{2}h^2) = -gh\partial_y b \end{cases} \tag{7}$$

where h is the water height, u and v are the fluid velocity components in orthogonal directions, b the bathymetry, g the gravitational acceleration and $\partial_{t,x,y}$ the partial derivatives with respect to time, and space in x and y directions, respectively.

The grid-based second-order FV method was implemented to solve the non-linear SW equations [11]. For the sake of consistency and simplicity, a description of the method for the 1D first order scheme is provided below, considering the flow in the x−direction. Denoting C_i as a cell with center x_i and length Δx, and assuming that all state variables U_i^n are known at time t^n, the FV method yields the values of the state variables at time t^{n+1} as given by Eq. (8):

$$U_i^{n+1} = U_i^n - \frac{\Delta t}{\Delta x}\left(F^n_{i+\frac{1}{2}} + \epsilon^n_{i+\frac{1}{2},L} - F^n_{i-\frac{1}{2}} - \epsilon^n_{i-\frac{1}{2},R}\right) + \Delta t S_i^n \tag{8}$$

where $F^n_{i+\frac{1}{2}}$ is the conservative flux across the cell interface $x_{i+\frac{1}{2}}$ with $U_i^n = (h_i^n, \eta_i^n, u_i^n)$ the vector of conservative variables. The source term S_i^n represents the discretization of the regular variations of the bathymetry. The non-conservative flux associated with the bathymetry discontinuity across the cell interfaces is $\epsilon^n_{i+\frac{1}{2},L}$ and $\epsilon^n_{i-\frac{1}{2},R}$, for approximations at the left- and right-hand side, respectively.

The full description of the implemented method, including the validation process of the code through analytical, laboratorial and field benchmarks can be found in [11,46]. In summary, the scheme is well-balanced to preserve the lake-at-rest configuration by applying a compensation between the flux term and the non-conservative contribution of the bathymetry. The scheme also handles dry/wet interfaces and correctly localizes the maximum run-off position. Second-order accuracy is achieved with local linear reconstructions, while robustness is guaranteed by using an *a priori* limiting procedure, MUSCL, or an *a posteriori* control technique, MOOD [46].

3.1.3. Built-in force computation: coupled FV-SPH methods

Bearing in mind that the FV method is a fast and efficient solver for static domains and regular flows as long as breaking waves or interaction with complex structures are not involved, together with the fact that SPH is a computationally expensive method yet capable of handling complex hydraulic phenomena and geometries, the coupling of the FV and SPH methods a compromise to obtain fast, yet accurate solutions.

The modeling technique makes use of the SPH and FV capacity to capture physical quantities of the fluid in a virtual position. By imposing Dirichlet boundary conditions inherent to each model, the interface where the data-exchange occurs are prescribed in a natural manner:

- the transition from SPH to FV is achieved by computing the hydrodynamic quantities at a virtual position coinciding with the location of the TBC used to initiate the FV domain. The SPH treats the quantities summing the relevant properties of all the particles inside the influence of the smoothing length, $2h_{sph} \approx 4\Delta p$, where Δp is the initial particle inter-distance. Then, the time–histories of the flow velocity are assigned in the horizontal direction, while the time–history of the free-surface elevation is imposed by assuming the height as a function of the highest particle position;
- the transition from FV to SPH is obtained by capturing the free-surface elevation and flow velocity at the cell that immediately precedes the position that coincides with the location of the OBC used to initiate the SPH domain. The OBC [75] is prescribed by a buffer layer of particles with a chosen width of $6\Delta p$ in the normal direction of the OBC.

3.2. Numerical modeling

To mimic the stages of the experiment, the whole domain has been divided into three zones: generation of the wave, followed by reflections and rarefactions, and ended with a slow energy decay, as depicted in Fig. 3. For the sake of coherence due to low-cost computation solutions and motivated by the experimental records, the influence of 2D and 3D models in the quality of the numerical solution is studied herein considering the 2D domain following the cut-line in the x−direction of the tank ($y = 1.83$ m).

Given the elevated geometry of the structure and symmetry of the experimental setup, the structure represents minimal influence in the lateral fluid flow, as corroborated by quantities of the physical experiment [47]. Acknowledging (1) the NS-SPH capacity to consider the gap between the free surface and the structure, modeling the structure as a partial obstacle to the flow, while (2) in the SW-FV simulations the hydrodynamic quantities are measured at the cell immediately before the presence of the structure, the hypothesis of reducing the numerical setup to a 2D was considered and calibrated against the experimental data, demonstrating that it is a reasonable assumption.

As illustrated in Fig. 3, the first modeling configuration considered makes use of the Lagrangian SPH method to solve the NS system of equations. The set of simulations was performed with the DualSPHysics software, running on a single graphics processing unit, GPU, card NVIDIA GeForce RTX 2060, @2.6 GHz, with 2 GB of memory, 1920 CUDA cores (64 CUDA cores per 30 multiprocessors). As one of SPH's drawbacks is the low rate of convergence, a brief convergence study on the 3D vs. 2D modeling assumption, using different Δp influencing the solutions of free surface elevation was performed to determine the admissible largest particle initial spacing that is capable of capturing the hydrodynamic flow conditions while controlling computational costs. Based on existing literature [76–78] a ratio of $\frac{H}{\Delta p} = 10$ was considered to set the values of Δp ranging from 10 cm to 1 cm and compared to a reference solution. The reference solution for the 2D models was obtained with $\Delta p = 0.005$ m, whereas for the 3D models and due to computational limitations, the reference solution was calculated

Table 1
Computational settings for the numerical simulations.

Lagrangian NS-SPH simulations	
Initial boundary condition	MBC and OBC
Kernel function	Wendland
Temporal discretization	Symplectic
CFL coefficient	0.2
Initial inter-particle distance	0.03 m
Coefficient for artificial viscosity	0.01
Reference fluid density	1100 kg/m^2
Time of simulation	150 s
Eulerian SW-FV simulations	
Initial boundary condition	TBC
Mesh discretization Δx	0.12 m
Mesh discretization Δy	0.15 m
Limiting technique	MUSCL van Leer
CFL coefficient	0.45
Time of simulation	150 s

with $\Delta p = 0.02$ m. The 3D effects, although more representative near the elevated structure and mostly in the broken wave case, were considered negligible. For the sake of conciseness, Fig. 4 depicts the convergence analysis regarding the 2D solutions for the unbroken wave case.

The 2D models computed with $\Delta p = 0.03$ m, for the unbroken wave case, i.e. $\frac{H}{\Delta p} = 23$, yielded results with a reasonable discrepancy between peak values (≈ 5%). For the broken wave case a $\Delta p = 0.04$ m ($\frac{H}{\Delta p} = 43$) was deemed sufficient to yield converged solutions. In addition, the results with the converged discretizations proved a reasonable compromise between the accuracy and the computational cost of the numerical solutions for the unbroken and broken wave cases, ranging from 35 to 70 h of simulation time for the 2D models and 850 to 2000 h for 3D models, respectively. In these analyses, the time domain of the simulation was set to $t_f = 150$ s, with time-steps $\Delta t = 0.05$ s to guarantee an approximation to Courant–Friedrichs–Lewy coefficient, $CFL = 0.20$. The initiation of the analysis was prescribed by a MBC, at $x = 0$ m, that mimics the piston motion as shown in Fig. 1-b.

The second modeling configuration adopted uses the Eulerian FV method to solve the non-linear SW equations. The set of simulations was carried out on a laptop with an AMD Ryzen5 2500U @1.8gHz CPU using around 27000 cells. The code to solve the SW equations uses a second-order approximation in space, with well-balanced fluxes, and the second-order Heun scheme in time [11,46]. A parametric study on the refinement of the structured mesh was performed using different cell sizes. For the benchmark simulations, $\Delta x = 0.12$ m and $\Delta y = 0.15$ m, in x−direction and y−direction, respectively. The simulation time is $t_f = 150$ s with time-steps around $\Delta t = 0.01$ s controlled by a $CFL = 0.45$ to guarantee the numerical stability. Due to the inherent limitation of the present FV discretization to deal with variable domains in time, the initial condition is prescribed by a TBC, at $x = 2$ m, using hydrodynamic quantities computed by SPH.

The third modeling configuration adopted aims to provide an efficient, yet simple and low-cost, alternative configuration to perform the simulation. Considering the simplified, but fast nature of SW-FV and the sophisticated, but computationally demanding, nature of NS-SPH, this configuration uses the inherent characteristics and boundary conditions of each scheme to perform the simulation. To avoid the piston-type wave generation, the left-most Eulerian SW-FV part of the simulation is initiated by a TBC prescribed using time–histories captured in the Lagrangian NS-SPH simulation, at $x = 2$ m. The simulation of the area where the structure is implemented is performed by the Lagrangian NS-SPH, initiated by an OBC prescribed using time–histories captured in the Eulerian SW-FV simulation, at $x = 36$ m. Table 1 summarizes the parameters of the numerical simulations.

4. Numerical–experimental benchmark

The mean value and standard deviation of the physical data recorded in the experimental campaign are used to draw correlations between the numerical simulation results and their experimental counterparts. Results are shown for the unbroken and broken wave cases in Figs. 5 and 6, respectively. Each figure includes a comparison of water elevation, flow velocity, horizontal and vertical forces and pressures exerted on the elevated structure, where the colorful curves represent the numerical solutions and the black lines represent the mean laboratorial data complemented by the standard deviation of the physical data derived from the trials that composed the experimental campaign, represented in shaded gray and identified as “Error”.

A reservation shall be made regarding the assessment of forces and pressures exerted vertically on the structure. When considering the semi-analytical approach, the hydrodynamic quantities are computed using the Eulerian SW-FV scheme, a depth-averaged approach. Due to the referred limitation, the vertical forces and pressures, in both unbroken and broken wave cases, are exclusively characterized by Lagrangian NS-SPH scheme (i.e. first and third numerical configurations).

4.1. Qualitative evaluation

From a qualitative perspective, the time-series of the numerical solutions fit the mean value and standard deviation of the physical data curves in most representations in Figs. 5 and 6.

As expected, among the experimental–numerical data correlation the NS-SPH approach is associated with the numerical solutions with better accuracy, while the SW-FV approach provides the smoother solutions and shows the inherent limitations to reproduce some phenomena, such as the characterization of vertical velocities (due to the SW simplification), not yielding results for the assessment of vertical force/pressure. The coupled Eulerian–Lagrangian modeling configuration provides numerical solutions with intermediate quality. The SW-FV scheme in the first part of the modeling configuration influences the overall quality of the solutions, capturing well the first peak, but not as well secondary peaks.

Capturing the vertical force induced by the broken wave proved the most challenging task, involving bore-like hydraulic phenomena and non-free surface, where a multi-phase air–water algorithm could possibly represent the difference in the accuracy of the solutions.

A noteworthy aspect to consider is the temporal window considered in the benchmark, $t = 150$ s. A smaller time interval focusing on the first wave would provide significantly improved values of agreement, but it would neglect two important aspects. First, hydrodynamic post-peak wave loading cycles impacting the structure, as recommended in ASCE 7 standard, which can increase adverse wave effects [4,79]. Second, the awareness for the cumulative artifacts of the numerical schemes, introducing deviations in the numerical approximation, become more noticeable after the first 50 s in Fig. 5. In Fig. 6, the latter is less evident due to a higher order of magnitude of the values.

From an overall appreciation of the modeling assumptions, the 2D modeling configurations with the numerical parameters summarized in Table 1 demonstrate their ability to reproduce the physics involved in the experimental setup.

4.2. Quantitative evaluation

From a quantitative perspective, a most conservative criterion is adopted to evaluate the fitting of the curves, considering exclusively the comparison between mean values of the recorded laboratorial data and the corresponding numerical solution for the total simulation time, $t =$ 150 s. The percentage of fitting discrepancy of the different experiment to numerical result correlations is set to the average percentage of the comparison between 300 points (time-steps of 0.5 s) and is calculated using the root mean square error approach. Table 2 lists the average

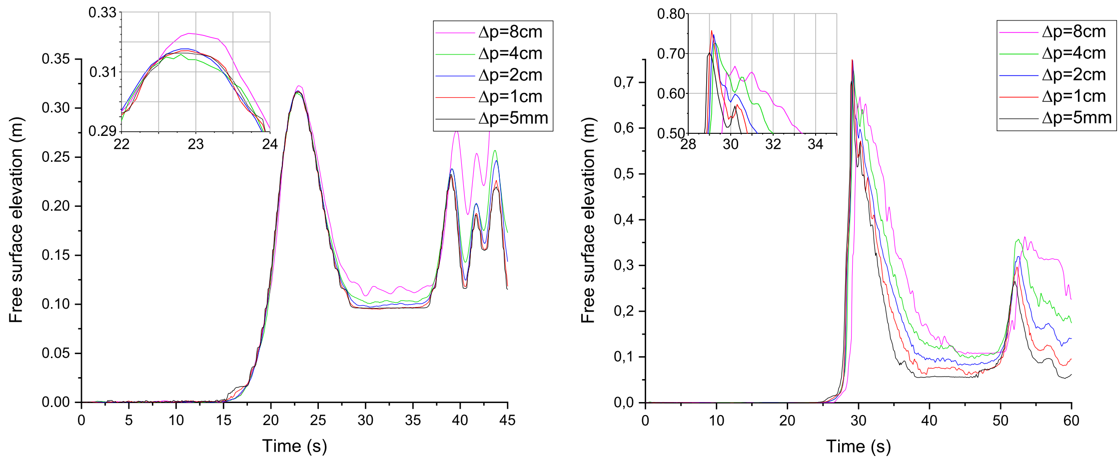


Fig. 4. Convergence analysis. Numerical solutions of the free surface elevation propagating over the flume (left) and near the structure (right).

Table 2
Benchmark results. Average disagreement between recorded physical quantities and numerical solutions with the time required to perform the simulations.

Runtime		30 h	<10 min	15 h
Simulation		Lagrangian	Eulerian	Coupled
Sensor	Case	Fitting disagreement [%]		
USWG6	unbr	2.50	4.62	3.57
	br	2.04	4.75	3.94
ADV7	unbr	7.73	10.48	9.27
	br	8.99	10.93	9.98
LC5	unbr	1.87	3.20	1.80
	br	1.50	1.68	0.72
LC6-9	unbr	4.25	–	10.54
	br	9.02	–	23.27
PG4	unbr	2.88	4.01	3.65
	br	0.94	0.73	1.22
PG11	unbr	3.65	–	9.69
	br	1.88	–	2.08

fitting disagreement. In the absence of other numerical validation criteria, the adequacy of the models is based on the reference value for the validation of tsunami inundation models, which require deviations below 10% [80,81].

Comparing in more detail the performance of the modeling configurations, the more immediate analytic assessments, using hydrodynamic quantities computed using the Eulerian SW-FV approach, provide general data on the magnitude of the load, while the more demanding Lagrangian NS-SPH approaches provide detailed aspects of the fluid impacting the structure, as expected. Additionally, the Lagrangian method has the capacity to overcome some of the numerical challenges, such as modeling a piston-type wave generator, and complement the Eulerian SW-FV in its inherent limitations. However, to compute the Lagrangian NS-SPH solution, a larger computational cost is required (running time of about 30 h for near 300000 particles with a multi-threaded GPU code) when compared to Eulerian simulations (less than 10 min to perform the same simulation, using a single core processor).

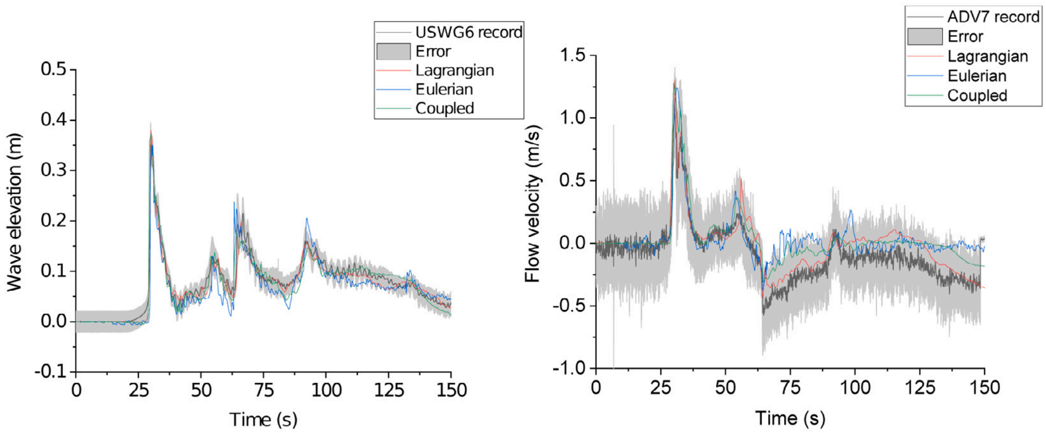
Taking advantage of the inherent features of each model-method, the coupled configuration represents a symbiotic relationship between Eulerian SW-FV and Lagrangian NS-SPH approaches. The computational effort is reduced when SW-FV computes the linear(ish) components of the model, leaving complex areas/zones, such as the fluid–structure interfaces or waves generated by pistons, to NS-SPH. By coupling the fast Eulerian and the sophisticated Lagrangian NS-SPH methods, the computation required about 15 h to perform a simulation with approximately 200000 particles. It is important to highlight the flexibility that the boundary conditions confer to assemble multiple numerical configurations besides the ones herein analyzed. By prescribing boundary conditions using the hydrodynamic quantities recorded by a laboratorial sensor or captured at a certain virtual gauge in SW-FV or NS-SPH simulations, different combinations of single or multiple domains can be derived. Such coupling procedures should be used with care, in adequacy to the problem it aims to simulate, bearing in mind they lead to trade-offs between accuracy and computational effort.

4.3. Discussion

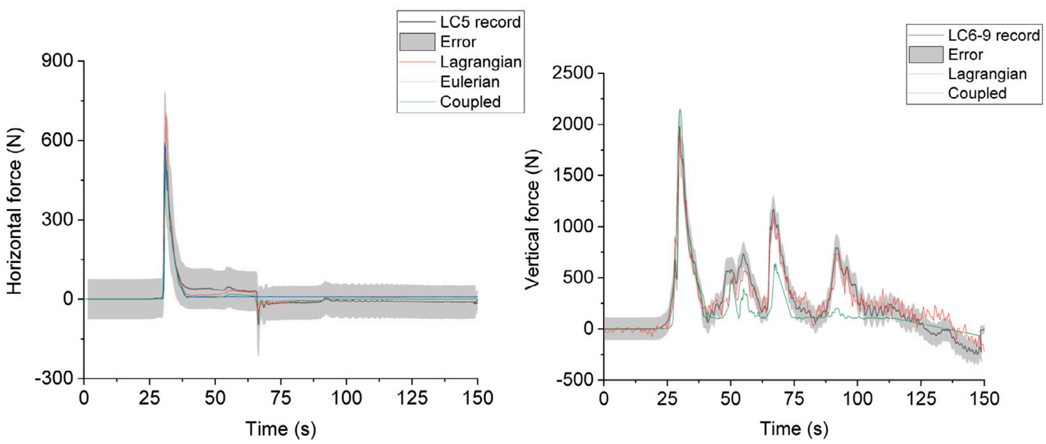
The interpretation of the results derive from the intrinsic characteristics of the governing equations and numerical schemes used. Due to the nature of the governing equations, the SW equations are an averaging approach of the NS equations with respect to the vertical coordinate, neglecting the viscosity effects. It is also worth mentioning that the SW finite volume code used here suffers from the limitation of disregarding the breaking wave in shallow water domain. Such a wave behavior frequently occurring when approaching the shoreline could have, as demonstrated in [12,82], a great importance in assessing the impact of long-wave phenomena (i.e, storm and tsunami) on coastal structures.

Moreover, the hyperbolic SW system presents some shock waves corresponding to breaking waves that are hardly reproduced by the model. For gravitational waves meeting the SW assumption, such a dissipation is quite low and the non-dissipative SW-FV system is eligible. On the other hand, turbulent flows require a large amount of viscous dissipation that cannot be directly produced using SW equations unless additional dissipation terms are introduced in the SW-FV tool, as the one used in the present benchmarking process, making possible the correct assessment of the flow behavior. On the contrary, the NS-SPH approach considers an empirical artificial viscosity term in its formulation, enabling the reproduction of the hydrodynamic effects, but the scheme tends to produce extra numerical diffusion that may be translated in a discrete overestimation of the physical viscosity. In the SPH simulation concerning the free surface for the broken wave case, it is possible to identify the referred phenomenon in the impact and reflection of the fluid on the structure from $t = 60$ s onward.

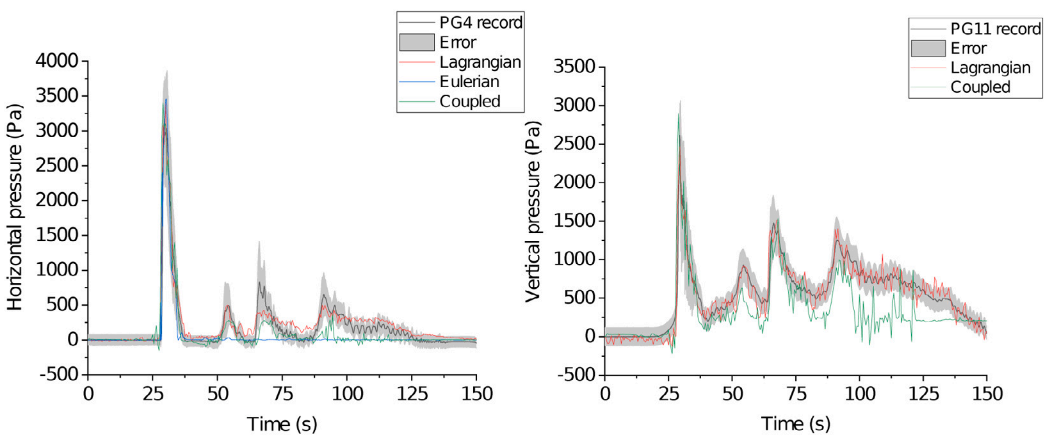
The numerical treatment of the elevated structure is constrained by the nature of Eulerian and Lagrangian methods. In the experimental three-dimensional (3D) setup, the structure is a partial occlusive obstacle corresponding to an elevated specimen, which allows water to flow under and around it, but in which, due to symmetry, the lateral flow has a negligible effect. In the 2D numerical models, the fluid is prevented from flowing around the structure, leading to slight over predictions of the quantities of interest, when compared to the range



(a) Hydrodynamic quantities: water elevation (left) and flow velocity (right).



(b) Force: horizontal (left) and vertical (right).

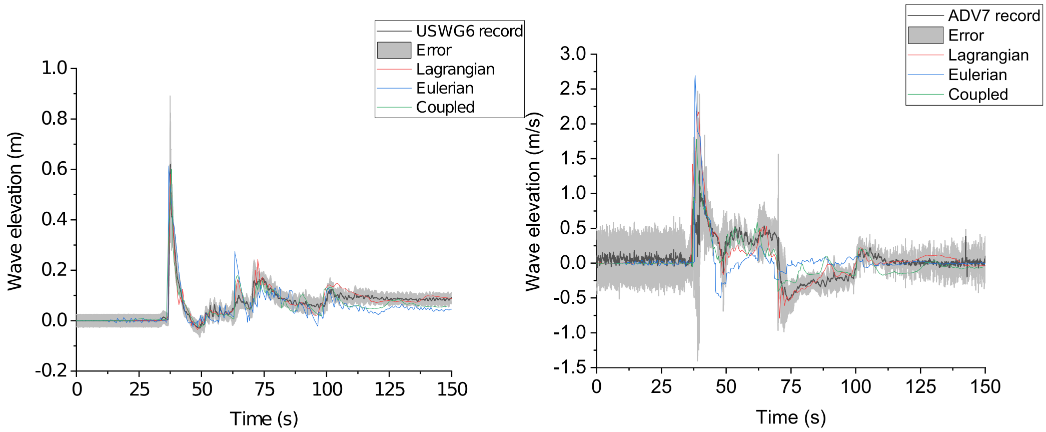


(c) Pressure: horizontal (left) and vertical (right).

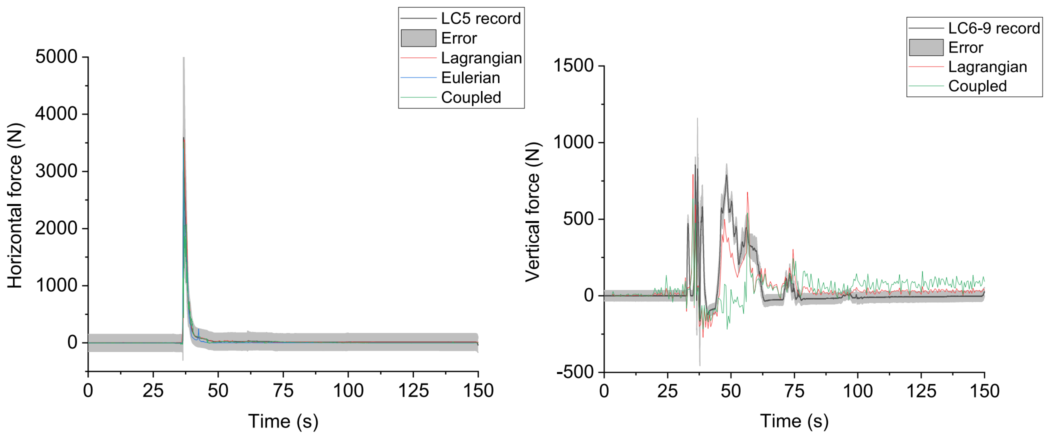
Fig. 5. Benchmarking for the unbroken wave case. Correlation between the mean ± standard deviation (error) laboratorial data and the numerical solution obtained using the Lagrangian NS-SPH, Eulerian SW-FV and coupled Eulerian–Lagrangian modeling configurations.

of the experimentally recorded data. The FV method, considering the structure as a z-elevation of a solid, assumes the structure to be a full obstacle with no space for the fluid to flow underneath, increasing the dam effect, as depicted in the correlations of the water elevation and velocity after the first 50 s. Less mass flow crossing the obstacle leads to an increase of reflected and retained fluid, respectively before and after the structure. The velocity profile is thus slightly reduced but with sharper shocks, higher amplitude and shorter duration. These observations are in accordance with the well-known feature of FV to treat simple free-surfaces with no breaking waves. On the contrary, the NS-SPH scheme models the space between the elevated structure and the flume based and walls into account, enabling the fluid to flow out numerically. This demonstrates the capacity of the NS-SPH model to reproduce the full geometry of the structure, reinforcing the acknowledgment of its adequacy to treat more complex free-surfaces and interaction with solids.

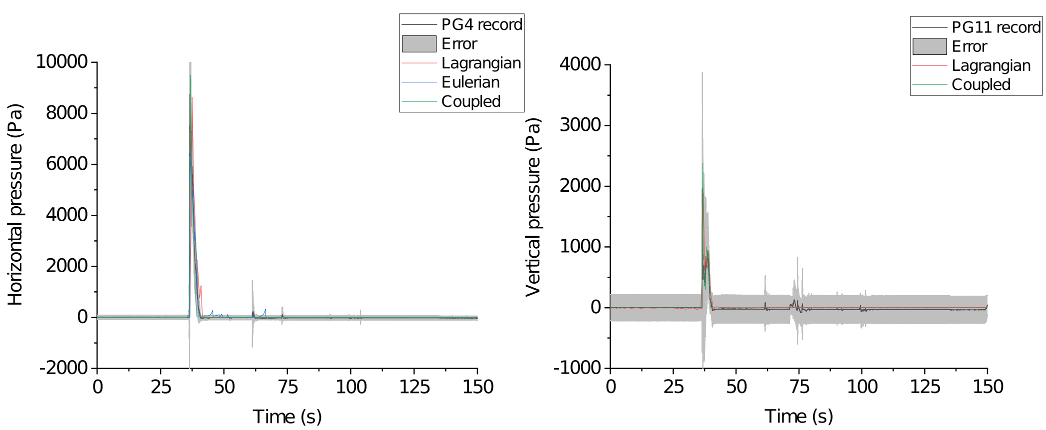
The solutions are also affected by the complexity of the modeling configuration, measured in terms of prescribed boundary conditions. Despite the complementary function of Lagrangian scheme to initiate characteristically fixed-domain in Eulerian simulation, each transition



(a) Hydrodynamic quantities: water elevation (left) and flow velocity (right).



(b) Force: horizontal (left) and vertical (right).



(c) Pressure: horizontal (left) and vertical (right).

Fig. 6. Benchmarking for the broken wave case. Correlation between the mean ± standard deviation (error) laboratorial data and the numerical solution obtained using the Lagrangian NS-SPH, Eulerian SW-FV and coupled Eulerian–Lagrangian modeling configurations.

from one model to another adds some deviations and an accumulation of numerical approximations, reducing the overall accuracy of the solution. The velocity in the vertical direction is exclusively quantified by the Lagrangian approach, i.e. the time series used to prescribe Dirichlet boundary condition in the different domains account exclusively for the free surface position and horizontal velocity.

Besides the intrinsic constrains posed by the nature of the numerical schemes influencing the quality of the final solutions, the characterization of force and pressure is also influenced by the numerical approach adopted to quantify them, built-in computation using NS-SPH scheme or analytic formulation using the hydrodynamic quantities computed by Eulerian SW-FV scheme. The latter Eulerian SW-FV approach introduces additional approximations in the quantification of force and pressure quantities due to (1) the semi-empirical nature of the hydrodynamic equation with the C_d value calibrated to match the peak value of the first wave, neglecting the time-dependency of C_d that leads to an under-estimation of the quantities after the first wave (Fig. 2), and (2) the position of the virtual gauges where the hydrodynamic quantities are measure is relocated for the cell that immediately precedes the one on which the fluid impacts the solid, to avoid the limitation of Eulerian SW-FV scheme to deal with fluids interacting with solids. Depending on the mesh discretization, such procedure can represent a difference of some centimeters with respect to the position occupied by the instruments in the experimental setup, representing additional uncertainty in the quantification of forces/pressures exerted on the structure.

5. Conclusions and future work

Experimental and numerical models are indispensable tools to study structures under tsunami action, but these can be extremely demanding limiting their adoption in current design practice. In this work, accessible numerical procedures were validated to contribute to the tsunami action characterization and, ultimately, improve assessment and design of tsunami-resistant structures.

The numerical solutions were compared with the laboratorial data acquired at an experimental campaign performed at the large wave flume of the Hinsdale Wave Research Laboratory, Oregon State University, on tsunami-like waves impacting an elevated structure.

The main conclusions from the correlation between physical and numerical quantities are:

- NS-SPH solutions are comparatively better than the SW-FV solutions, while coupling both numerical schemes represents a good compromise.
- The correlation between the numerical solutions and the recorded experimental data supports: (1) the hypothesis of converting the 3D experimental setup into a 2D numerical domain for the benchmark modeling performed, (2) the modeling configurations adopted in terms of spatial discretization, boundary conditions initiating and coupling the different sub-domains and respective location of implementation, (3) the SPH numerical assessment of forces and pressures exerted on the elevated structure, and (4) the semi-analytic quantification is particularly suited to assess the forces induced by the first wave.
- Based on the characterization of the drag coefficient when evaluating hydrodynamic forces C_d is very sensitive to minimal geometry or flow variations. For the investigated geometry and flow conditions, the estimated C_d values, for both the horizontal and vertical directions, are lower than the provisions from the ASCE 7 standard and the FEMA guidelines, which implies that both regulations would recommend conservative values to design the elevated structure. By validating the numerical solutions with experimental data, a numerical alternative to the more expensive and time-consuming experimental approach is provided, allowing to simulate numerous scenarios to help characterizing C_d. The Eulerian scheme can provide valuable insights on the numerical calibration of C_d in the horizontal direction since SW is a depth-average integration method, while the Lagrangian one is particularly more interesting since it can provide calibration insights on both, horizontal and vertical, directions.
- The quantitative evaluation of the schemes performance shows values of fitting disagreement between the experimental and numerical solutions up to 9% for the Lagrangian NS-SPH solutions, and up to 11% for the Eulerian SW-FV solutions. The coupled FV-SPH numerical configuration shows values bounded by the two former modeling schemes.
- The three numerical configurations adopted to reproduce the physics of the experimental campaign provide slightly better approximations for the unbroken wave than for the broken wave case highlighting the varying levels of performance of the modeling schemes when subjected to different flow conditions.
- The trade-offs between quality and accuracy of the solutions and computational effort were discussed, whereby the ratio of computational demand is about $1/180$ between Eulerian SW-FV and Lagrangian NS-SPH approaches, while the fitting disagreement was about $1/2$. The coupling of the numerical schemes reducing the SPH domain to the highly non-linear regions presented itself as a promising approach to balance the computational costs and keep the accuracy of the numerical solutions [41–45,83]. In the present setup, the reduction of the SPH domain to less than a half represented about 50% of computation time saving. In real coastal engineering problems, the scale of the tsunami propagation phase is much larger than the inundation and fluid-interaction phases of the experimental campaign, leading to a greater expression of the advantages of coupling.
- The complementary relation between numerical schemes is herein highlighted. The Lagrangian NS-SPH is able to handle complex and realistic geometries, but requires considerable computational efforts. The SW-FV is definitively faster, but unable to compute complex flows. Situations such as turbulent flows, as well as the ability to model piston-type wave-makers and the gaps between the water level and an elevated structure cannot be carried out with such a simple method.

As future work, a comparison between the 3D and 2D modeling would provide a more solid validation on the effect of neglecting the third coordinate when modeling physical setups with symmetric geometry and flow conditions. Nonetheless, to preserve the quality of the solutions, the Lagrangian NS-SPH numerical scheme would require a discretization into millions of particles, implying the use of high computational performance that can be carried out only in specific computer centers, whereas the simulations presented in the present work were performed using an accessible local cluster. Alternatively, parametric studies to correlate the accuracy of 2D simulations and potential factors that can influence the results could be performed, including wave directivity, structural orientation with respect to the wave propagation, geometry of the solid, among others, to provide important insights on the characterization of hydraulic phenomena.

Another aspect to consider in future works is a refined spatial discretization of the Eulerian models. A multi-grid feature with successive refinement is being currently implemented in the SW-FV tool. This feature allows to produce more accurate numerical solutions without adding a significant computational effort. The Eulerian SW-FV scheme would also benefit from the implementation of an algorithm of numerical determination of the pressure through the SW-FV scheme. Making use of the non-conservative flux terms, and by assessing the wave height and flow velocity at the cell immediately before the structure, it is possible to quantify the pressure. The velocity tends to zero at the instant of the impact, which means that the pressure turns to hydrostatic, allowing to apply the Bernoulli principles to determine an equivalent pressure.

CRediT authorship contribution statement

Cláudia Reis: Conceptualization, Methodology, Validation, Formal analysis, Writing – original draft, Funding acquisition. **Stéphane Clain:** Software, Writing – review & editing, Funding acquisition. **Jorge Figueiredo:** Software, Writing – review & editing. **André R. Barbosa:** Laboratorial resources, Analysis, Writing – review & editing. **Maria Ana Baptista:** Writing – review & editing. **Mário Lopes:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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